

HOPSTEN, Germany

WMO: 103140

Lat: 52.333N

Lon: 7.533E

Elev: 131

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 90-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.1	(c) 19.4	(d) 11.1	(e) 9.7	(f) 16.4	(g) 15.4	(h) 12.1	(i) 22.2	(j) 31.1	(k) 46.6	(l) 27.5	(m) 46.4	(n) 6.3	(o) 70	(p) 0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.5	(c) 86.6	(d) 67.1	(e) 82.7	(f) 65.3	(g) 78.7	(h) 63.5	(i) 69.1	(j) 82.2	(k) 66.8	(l) 78.5	(m) 65.0	(n) 75.6	(o) 6.8	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	91.0	73.1	62.6	85.2	71.5	60.9	80.2	69.2	33.2	82.3	31.4	78.4	30.0	76.2	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.5	(b) 19.3	(c) 16.9	(e) 9.7	(f) 92.6	(g) 7.7	(h) 2.7	(i) 4.1	(j) 94.6	(k) -0.4	(l) 96.2	(m) -4.7	(n) 97.7	(o) -10.3	(p) 99.6		
			DB	WB	9.4	72.3	7.5	1.9	4.0	73.7	-0.4	74.8	-4.6	75.9	-10.1	77.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	36.8	38.4	43.0	48.0	56.0	59.9	64.5	64.4	57.7	50.7	42.3	36.7		
	DBStd	12.10	8.44	8.44	5.75	6.98	6.98	6.04	5.85	5.61	4.87	6.39	6.37	8.06		
	HDD50	1833	410	327	226	118	22	1	0	0	2	73	238	416		
	HDD65	5666	873	744	682	510	289	177	80	80	228	444	682	877		
	CDD50	1812	2	3	9	60	209	298	450	448	231	93	6	3		
	CDD65	171	0	0	0	1	11	24	65	63	7	0	0	0		
	CDH74	2070	0	0	0	18	176	321	755	720	79	1	0	0		
	CDH80	646	0	0	0	0	30	92	251	255	18	0	0	0		
(14) Wind	WSAvg	7.4	9.1	9.3	8.6	7.4	6.8	6.4	6.1	5.9	6.2	7.2	7.4	8.1		
(15) Precipitation	PrecAvg	32.7	3.0	2.5	2.4	1.9	2.3	2.7	3.2	2.9	3.0	2.7	2.9	3.0		
	PrecMax	41.8	5.2	5.3	5.1	3.8	5.0	5.1	6.6	5.3	6.1	8.6	4.9	6.9		
	PrecMin	25.6	0.3	0.8	0.5	0.2	1.2	1.0	1.3	0.8	0.4	0.6	1.0	1.0		
	PrecStd	5.0	1.5	1.3	1.3	0.9	0.9	1.1	1.5	1.2	1.5	1.7	1.2	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	59.2	64.6	77.0	84.1	88.3	91.1	91.3	82.4	71.7	58.6	55.8		
		MCWB	50.6	50.4	52.2	58.1	63.6	69.1	69.8	67.6	64.0	61.5	52.9	52.2		
	2%	DB	52.2	54.8	59.2	69.7	78.8	82.2	86.4	86.6	74.8	66.4	55.5	53.6		
		MCWB	48.2	48.4	50.0	56.1	61.4	65.3	67.9	66.7	62.8	58.8	52.6	50.9		
	5%	DB	50.1	52.0	55.3	64.9	74.9	77.1	82.3	82.3	69.5	62.7	53.1	50.4		
		MCWB	46.8	47.5	48.6	53.1	59.7	63.6	66.0	65.3	60.3	57.4	49.6	47.5		
	10%	DB	47.6	49.3	52.3	60.2	70.0	72.7	77.6	77.1	66.2	59.9	50.4	47.9		
		MCWB	44.9	45.6	47.1	51.1	57.8	61.4	63.5	63.9	59.0	55.4	47.6	45.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	51.4	53.8	60.3	64.6	70.3	71.4	71.6	66.8	62.7	54.9	53.3		
		MCDB	54.5	56.5	61.1	72.3	79.8	83.5	88.3	86.4	79.7	68.5	56.8	55.2		
	2%	WB	49.2	49.6	51.4	57.5	62.7	67.0	69.2	68.7	63.5	59.8	52.5	50.6		
		MCDB	51.4	53.7	56.8	67.4	75.6	79.2	83.1	81.3	71.8	64.7	55.1	53.4		
	5%	WB	47.1	47.7	49.7	54.8	60.4	64.4	67.0	66.4	61.7	57.7	50.2	47.9		
		MCDB	49.9	51.5	54.1	62.5	71.7	73.8	78.6	77.6	67.9	61.9	52.4	50.1		
	10%	WB	44.5	45.7	47.9	52.1	58.4	62.1	64.8	64.8	60.0	55.8	48.1	45.2		
		MCDB	47.0	49.0	52.1	58.8	68.2	70.8	75.0	75.3	64.9	59.6	50.1	47.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	10.9	13.5	17.5	19.3	18.5	19.5	20.3	16.3	14.1	10.4	8.5		
		MCDBR	11.3	13.8	18.9	24.2	26.8	26.9	27.9	29.2	22.0	17.5	12.3	10.6		
		MCWBR	9.3	10.0	12.0	12.9	12.6	13.4	12.3	12.8	12.3	11.4	9.5	9.1		
	5% WB	MCDBR	10.3	12.2	15.9	21.2	23.3	23.9	25.1	24.5	18.7	16.5	11.6	10.3		
		MCWBR	9.2	9.7	11.1	11.9	12.0	13.1	11.6	12.2	11.5	11.4	9.6	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.346	0.376	0.410	0.412	0.410	0.422	0.415	0.389	0.361	0.341	0.319		
	taud		2.357	2.340	2.269	2.218	2.247	2.271	2.248	2.287	2.350	2.409	2.373	2.342		
	Ebn at Noon		210	241	256	261	266	268	262	257	249	233	202	190		
	Edh at Noon		19	26	34	40	41	40	41	37	31	24	19	17		
(44) All-Sky Solar Radiation	RadAvg		215	429	785	1274	1553	1649	1580	1328	965	562	258	168		
	RadStd		25	75	107	138	139	127	200	112	112	55	28	18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page